

1. Administrative & Study Identification

Full Study Title: An Open-label, Multicenter Study to Evaluate Pharmacokinetics, Safety and Tolerability up to 6 Years of Intravenous Secukinumab Infusions in Pediatric Participants With Juvenile Psoriatic Arthritis **Short Title/Acronym:** Study to Evaluate the PK, Safety and Tolerability of i.v. Secukinumab in Pediatric Participants With JPsA **Protocol Number:** CAIN457G22101 **NCT Number:** NCT06751238 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Secukinumab (Intravenous / i.v.) / Trade Name: Cosentyx (ATC Code: L04AC10) **Phase of Development:** Phase 1 **Trial Status:** Recruiting **Medical Monitor:** Novartis Pharmaceuticals Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the pharmacokinetics (PK), safety, and tolerability of multiple doses of intravenous (i.v.) secukinumab in pediatric participants with Juvenile Psoriatic Arthritis (JPsA). **Secondary Objectives:** To assess the number of participants experiencing Adverse Events (AEs), Serious Adverse Events (SAEs), and clinically significant changes in clinical laboratory measures and vital signs up to Week 20.

2.2 Background & Rationale To address the clinical need for effective, long-term biologic therapies in the pediatric population by evaluating the PK profile, safety, and tolerability of an i.v. formulation of secukinumab over a period of up to 6 years. This study targets pediatric participants who have demonstrated an inadequate response or intolerance to traditional Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) or Disease Modifying Anti-Rheumatic Drugs (DMARDs).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target \$N\$: 20 participants **Number of Sites:** Multicenter **Estimated Study Start:** 24-Sep-2025 **Estimated Primary Completion:** 30-Nov-2032

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participant's parent's or legal representative(s) written informed consent and child's assent, if appropriate, must be obtained before any study-related activity or assessment is performed. If the participant reaches the age of consent (as per local law) during the study, they must also sign the corresponding study Informed Consent Form (ICF). 2. Males and females ≥ 2 years old to < 18 years old at the time of screening. 3. Confirmed diagnosis of JPsA according to the modified International League of Associations for Rheumatology (ILAR) classification criteria that must have occurred at least 6 months prior to screening. 4. Active JPsA disease defined as ≥ 3 active joints (swollen, or if not swollen must be both tender and limited range of motion) at baseline (BSL). 5. Inadequate response (≥ 1 month) or intolerance to ≥ 1 NSAID at screening. 6. Inadequate response (≥ 2 months) or intolerance to ≥ 1 DMARD at screening.

4.2 Exclusion Criteria 1. Participants with a body weight less than 10 kg at screening. 2. Participants with active inflammatory bowel disease or active uveitis at screening or baseline. 3. Use of other investigational drugs within 4 weeks or 5 half-lives of BSL, or until the expected pharmacodynamic effect has returned to BSL, whichever is longer. 4. History of hypersensitivity to the study drug or its excipients or to drugs of similar chemical classes. 5. Fulfilling diagnostic criteria for any ILAR juvenile idiopathic arthritis (JIA) category other than JPsA at BSL. 6. Participants treated with prohibited medication. 7. Participants taking any non-biologic DMARD at screening except for MTX. 8. Any medical or psychiatric condition which, in the investigator's opinion, would preclude the participant from adhering to the protocol or completing the study per protocol.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple doses of secukinumab administered during the study and optional extension period. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Up to 6 years.

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant use of second-line agents is allowed, including: stable doses of methotrexate (MTX) (maximum of $20 \text{ mg/m}^2 \text{ BSA/week}$) for at least 4 weeks prior to BSL with folic/folinic acid supplementation; an oral corticosteroid (CS) at a prednisone equivalent dose of $< 0.2 \text{ mg/kg/day}$ or up to 10 mg/day maximum (whichever is less) for at least 7 days prior to BSL; and stable dose of no more than one NSAID for at least 1 week prior to BSL. **Prohibited:** Any non-biologic DMARD at screening except for MTX, use of other investigational drugs, and any other prohibited medications as defined by the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Prior to Day 0 Informed Consent/ Assent, Medical History, Vital Signs, Weight check
2	Baseline	Day 0 First IV Infusion, PK baseline sampling, Joint evaluation
3	Interim	Up to Week 20 & Beyond PK sampling, Safety Labs, AE/SAE monitoring, Efficacy monitoring
4	End of Study	Year 6 (Optional Ext.) Final PK and Safety assessments, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Evaluation of pharmacokinetics (PK), safety, and tolerability parameters of i.v. secukinumab. Specific primary outcome measures are: 1. Maximum concentration of secukinumab on Day 1. 2. Maximum concentration at steady-state (C_{max}, ss). 3. Minimum concentration at steady-state (C_{min}, ss). **Measurement Method:** Serial pharmacokinetic blood sampling and clinical safety assessments.

7.2 Secondary Endpoints 1. Number of participants with Adverse Events (AEs) and Serious Adverse Events (SAEs). 2. Number of participants with clinically significant changes in clinical laboratory measures and vital signs.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via physical examinations, vital signs, and laboratory measures up to Week 20 and throughout the 6-year extension. **Serious Adverse Events (SAEs):** Continuously monitored and reported per sponsor and regulatory protocols. **Data Monitoring Committee (DMC):** Internal/ External safety monitoring managed by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE/SAE evaluations; PK Analysis Set for pharmacokinetic modeling. **Sample Size Justification:** Target N=20, standard for pediatric Phase 1 PK, safety, and tolerability studies. **Handling of Missing Data:** Standard sponsor imputation protocols (e.g., LOCF or Multiple Imputation) for PK concentration-time profiles and safety analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring visits ensuring adherence to the 6-year treatment extension protocol.

Audit Trail: Verification of electronic Case Report Forms (eCRF) and data clarification protocols maintained per Novartis standards.

11. Study Identifiers & Governance

Primary ID: CAIN457G22101 **Registry Number:** NCT06751238 **Sponsor/Lead Organization:** Novartis **Study Title:** Study to Evaluate the Pharmacokinetics (PK), Safety and Tolerability up to 6 Years of Intravenous (i.v.) Secukinumab in Pediatric Participants With Juvenile Psoriatic Arthritis (JPSA) **Official Regulatory Title:** An Open-label, Multicenter Study to Evaluate Pharmacokinetics, Safety and Tolerability up to 6 Years of Intravenous Secukinumab Infusions in Pediatric Participants With Juvenile Psoriatic Arthritis

12. Clinical Framework (Juvenile Psoriatic Arthritis Specific)

Primary Condition: Juvenile Psoriatic Arthritis (JPSA) **Diagnosis Threshold:** Confirmed JPSA (modified ILAR criteria) with ≥ 3 active joints (swollen/tender with limited range of motion)

Medical Methodology: Biologic Disease-Modifying Anti-Rheumatic Drug (Targeted IL-17A Inhibitor Therapy) **Optimization Target:** Establishment of pediatric IV dosing PK profile and long-term safety

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for pediatric JPSA utilizing biologic intervention **Education Protocol (HCP):** Site initiation and continuous training on pediatric PK sampling and IV administration **Education Protocol (Patient):** Administration of Assent and parental Informed Consent; caregiver training on JPSA management and IV infusion protocol **Quality Audit Mechanism:** Clinical monitoring of PK sample integrity and patient AE/SAE adherence tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 72 months (6 Years) **Onsite Visit Cadence:** Baseline, frequent intervals up to Week 20, and extended

continuous maintenance phases **Remote Monitoring Frequency:** As specified by the trial's clinical operations plan **Checklist Review Interval:** Regular assessment of concomitant medications, joint counts, and emergent AEs

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				